

# Transmission Network Reduction

We merge buses so that a DC-OPF on the smaller network still gives a dispatch that works on the original one. All three approaches choose the same thing — which lines to collapse — and differ in what they require of the result.

## 1 Notation

|                                         |                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------|
| $\mathcal{B}, \mathcal{L}, \mathcal{G}$ | buses 1.. $N$ , lines 1.. $L$ , generators 1.. $K$                            |
| $u(l), v(l)$                            | ends of line $l$ ; $n(k)$ the bus of generator $k$                            |
| $E$                                     | bus–line incidence: $E_{bl} = +1$ if $u(l) = b$ , $-1$ if $v(l) = b$ , else 0 |
| $C$                                     | bus–generator incidence: $C_{bk} = 1$ if $n(k) = b$ , else 0                  |
| $\mathbf{B}$                            | $\text{diag}(B_1, \dots, B_L)$ , the line susceptances                        |
| $F_l$                                   | thermal rating of line $l$                                                    |
| $\underline{T}_l, \overline{T}_l$       | bounds on the transfer a collapsed line may carry                             |
| $c_k$                                   | linear cost of generator $k$ ; $\underline{g}_k, \overline{g}_k$ its limits   |
| $d$                                     | demand at each bus; $p = Cg - d$ the nodal injections                         |
| $g, \theta, f$                          | generation, bus angles, line flows                                            |
| $t$                                     | internal transfer, carried only by a collapsed line                           |
| $z_l$                                   | 1 if line $l$ is collapsed, 0 if it survives                                  |
| $Z^*$                                   | optimal cost of the full network (a constant)                                 |

## 2 What a reduction is

Both MILP models express the reduction on the original graph, using one switch:

$$-F_l(1 - z_l) \leq f_l \leq F_l(1 - z_l) \quad \text{switched limit} \quad (1)$$

At  $z_l = 0$  this is the ordinary limit. At  $z_l = 1$  it forces  $f_l = 0$ , hence  $\theta_{u(l)} = \theta_{v(l)}$ : the buses merge. The power the line carries is then taken up by the internal transfer  $t_l$ , which is free only on a collapsed line. So a collapsed line loses its thermal limit, which is the whole difficulty.

With the transfer present the DC power flow reads  $E \mathbf{B} E^\top \theta + Et = Cg - d$ .

### 3 Flow matching

The **proxy/** approach. Fix the injections at the values the full OPF produced and ask the reduced network to reproduce the same flows to within a tolerance. No optimisation sits inside the model, so it is one MILP. Here  $\hat{f}$  and  $p$  are the full network's optimal flows and injections, and  $\varepsilon_l$  is the window half-width (**eps** times the rating). Lines loaded above **near\_limit** are held out of any cluster.

$$\max_{z, \theta, f, t} \sum_{l \in \mathcal{L}} z_l \quad \text{maximise merges} \quad (2)$$

$$f = \mathbf{B} E^\top \theta \quad \text{line flows} \quad (3)$$

$$E(f + t) = p \quad \text{DC power flow, fixed injections} \quad (4)$$

$$(\hat{f}_l - \varepsilon_l)(1 - z_l) \leq f_l \leq (\hat{f}_l + \varepsilon_l)(1 - z_l) \quad \text{flow window, switched} \quad (5)$$

$$\underline{T}_l z_l \leq t_l \leq \overline{T}_l z_l \quad \text{internal transfer} \quad (6)$$

$$z_l = 0, \quad l \text{ congested} \quad \text{line stays unmerged} \quad (7)$$

for all  $l \in \mathcal{L}$ .

The injections are fixed, so accuracy is asserted only at the dispatch the full network already chose. A DC-OPF on the reduced network may pick a different one, and that dispatch can overload the original network.

## 4 Bilevel

Ask for what flow matching cannot: the dispatch the *reduced* network returns must be deliverable on the original one. Here  $\tilde{\theta}, \tilde{f}$  are the angles and flows the original network carries under that dispatch.

$$\max_{z \in \{0,1\}^L} \sum_{l \in \mathcal{L}} z_l \quad \text{maximise merges} \quad (8)$$

$$\tilde{f} = \mathbf{B} E^\top \tilde{\theta}, \quad |\tilde{f}_l| \leq F_l \quad \text{deliverable on full network} \quad (9)$$

$$E \mathbf{B} E^\top \tilde{\theta} = Cg - d \quad \text{DC power flow, full network} \quad (10)$$

$$c^\top g \leq (1 + \frac{\gamma}{100}) Z^\star \quad \text{cost within tolerance} \quad (11)$$

$$(g, \theta, f, t) \in \arg \min \text{LL}(z) \quad \text{lower level optimal} \quad (12)$$

The lower level  $\text{LL}(z)$  is the reduced network's DC-OPF:

$$\min_{g, \theta, f, t} c^\top g \quad \text{minimise cost} \quad (13)$$

$$f = \mathbf{B} E^\top \theta \quad \text{line flows} \quad (14)$$

$$E(f + t) = Cg - d \quad \text{DC power flow} \quad (15)$$

$$-F_l(1 - z_l) \leq f_l \leq F_l(1 - z_l) \quad \text{switched limit} \quad (16)$$

$$\underline{T}_l z_l \leq t_l \leq \overline{T}_l z_l \quad \text{internal transfer} \quad (17)$$

$$\underline{g}_k \leq g_k \leq \overline{g}_k \quad \text{generator limits} \quad (18)$$

### 4.1 Single-level KKT form

$\text{LL}(z)$  is an LP for fixed  $z$ , so (12) can be replaced by its KKT conditions. The single-level model is

(8)–(11) (upper level), (13)–(18) (lower-level feasibility), and the rows below.

Duals are  $\lambda$  on (15),  $\sigma$  on (14),  $\mu^\pm \geq 0$  on (16),  $\tau^\pm \geq 0$  on (17), and  $\rho^\pm \geq 0$  on (18).

$$c - C^\top \lambda + \rho^+ - \rho^- = 0 \quad (g) \quad (19)$$

$$E^\top \lambda + \sigma + \mu^+ - \mu^- = 0 \quad (f) \quad (20)$$

$$E^\top \lambda + \tau^+ - \tau^- = 0 \quad (t) \quad (21)$$

$$E \mathbf{B} \sigma = 0 \quad (\theta) \quad (22)$$

$$\mu_l^+ (F_l(1 - z_l) - f_l) = 0 \quad \text{complementarity, flow upper} \quad (23)$$

$$\mu_l^- (F_l(1 - z_l) + f_l) = 0 \quad \text{complementarity, flow lower} \quad (24)$$

$$\tau_l^+ (\overline{T}_l z_l - t_l) = 0 \quad \text{complementarity, transfer upper} \quad (25)$$

$$\tau_l^- (t_l - \underline{T}_l z_l) = 0 \quad \text{complementarity, transfer lower} \quad (26)$$

$$\rho_k^+ (\overline{g}_k - g_k) = 0, \quad \rho_k^- (g_k - \underline{g}_k) = 0 \quad \text{complementarity, generators} \quad (27)$$

## 5 Greedy

Merge one line at a time, accepting a merge only if two LPs confirm the same condition the bilevel model asks for. Both LPs carry the reduced and the original network at once, sharing the dispatch  $g$ .

$$\phi(z) = \min c^\top g \text{ s.t. reduced network} \quad \text{reduced optimum} \quad (28)$$

$$\eta(z) = \min \eta \text{ s.t. } |\tilde{f}_l| \leq F_l(1 + \eta), \ c^\top g \leq \phi(z) + \epsilon_c \text{ cheapest violation} \quad (29)$$

$$\phi(z) \leq (1 + \frac{\gamma}{100})Z^* \quad \text{accept if: cost within tolerance} \quad (30)$$

$$\eta(z) \leq \epsilon_f \quad \text{accept if: no overload} \quad (31)$$

and the same must hold at every design demand. Equation (29) searches the reduced network's optimal cost band for a dispatch safe on the original network, which is the “some optimum” of (12).

Candidates are ordered by how much a merge moves the flows. Collapsing line  $(i, j)$  at fixed injections shifts every flow by

$$\Delta f = \kappa (H_{:,i} - H_{:,j}), \quad \kappa = -\frac{\theta_i - \theta_j}{X_{ii} + X_{jj} - 2X_{ij}} \quad \text{predicted flow change} \quad (32)$$

$$\text{score} = \alpha \|\Delta f/w\|_2 + (1 - \alpha) \|\Delta f/w\|_\infty \quad \text{ranking score} \quad (33)$$

with  $H$  the PTDF,  $X$  the reference-removed inverse of  $E\mathbf{B}E^\top$ , and  $w$  the normaliser: either the ratings  $F$  or the remaining headroom  $F - |f|$ , set by `norm`. This is exact; after each accepted merge  $X$  and  $H$  are updated by rank-one formulas.

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### Algorithm 1 Greedy reduction

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- 1:  $z \leftarrow 0$ ; check the full network passes the acceptance test
  - 2: merge every radial, generator-free line without an LP
  - 3: build  $X$ ,  $H$  and score all candidates
  - 4: **while** time remains and candidates are left **do**
  - 5:    $l \leftarrow$  lowest-scoring candidate
  - 6:    $z' \leftarrow z$  with  $l$ , and every line its clustering makes internal, set to 1
  - 7:   **if**  $z'$  passes the acceptance test at every design demand **then**
  - 8:      $z \leftarrow z'$ ; update  $X$ ,  $H$ ; re-score
  - 9:   **else**
  - 10:     drop  $l$
  - 11:   **end if**
  - 12: **end while**
  - 13: **optional:** solve the joint KKT model at fixed  $z$ ; on failure roll back and repeat
  - 14: **return**  $z$
- 

Every accepted merge satisfies the bilevel condition. Nothing says the reduction is the largest possible: there is no dual bound, and merges are irreversible, so it can stall while a better reduction exists that needs two changes at once.

## 6 Summary

|               | injections                | guarantee                              |
|---------------|---------------------------|----------------------------------------|
| Flow matching | fixed at the full optimum | none on the re-optimised dispatch      |
| Bilevel       | chosen by the reduced OPF | some reduced optimum is safe and cheap |
| Greedy        | chosen by the reduced OPF | same, per accepted merge               |

All three certify only the demands in the design set, and only for the lossless DC model.